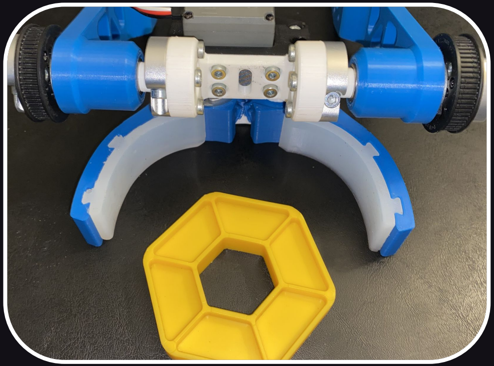
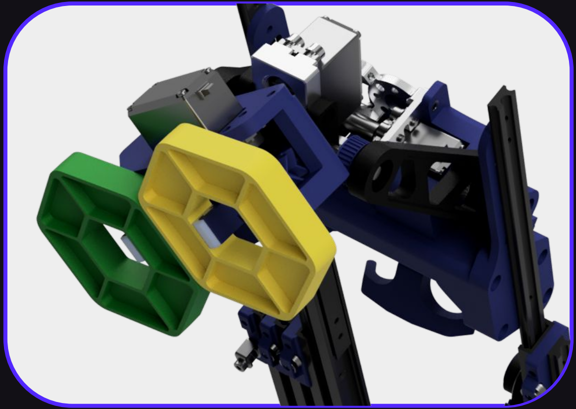
2023-2024 Season Robot

~Video~

The challenge for this season included intaking and orienting different colored hexagon game pieces. This robot would passively intake the game pieces from the left side and outtake them / orient them on the right side.

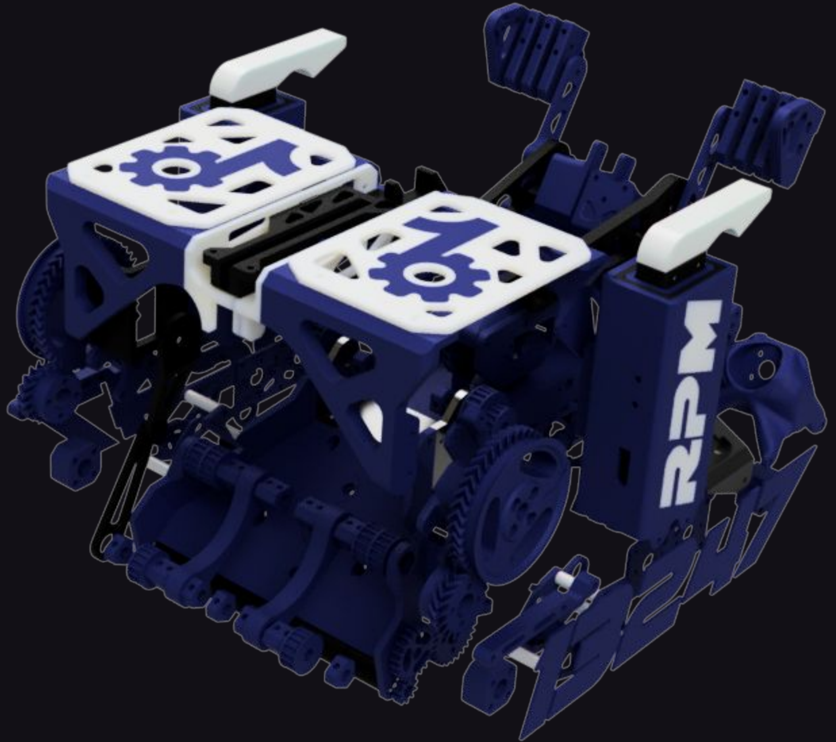


In the original design, the robot would intake the game pieces with a claw. This ended up being too slow which drove our decision to switch to a passive noodle-based intake

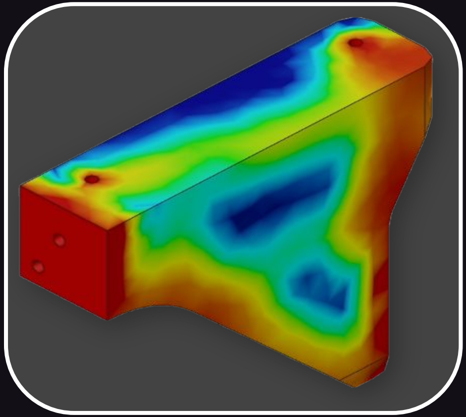


The outtake of the robot grabbed the game pieces by the middle and could rotate 300 degrees for any type of orientation necessary.

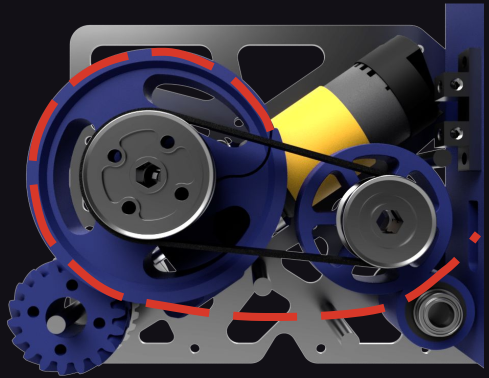
2023-2024 Season Robot



This image shows every 3D printed part on our robot this season. This robot ended with a total of 160 3D printed parts from multiple filaments like PLA, ABS, ASA, and TPU.



One issue with this robot was reducing weight, to solve this, we utilized topology optimization to find where to remove material without sacrificing structure.



At the end of the match the robot could score points by climbing a bar 2 ft off of the ground. To do this, our robot used a spool and winch system to push telescoping 3D printed slides to the height

2022-2023 Season Robot

~Video~

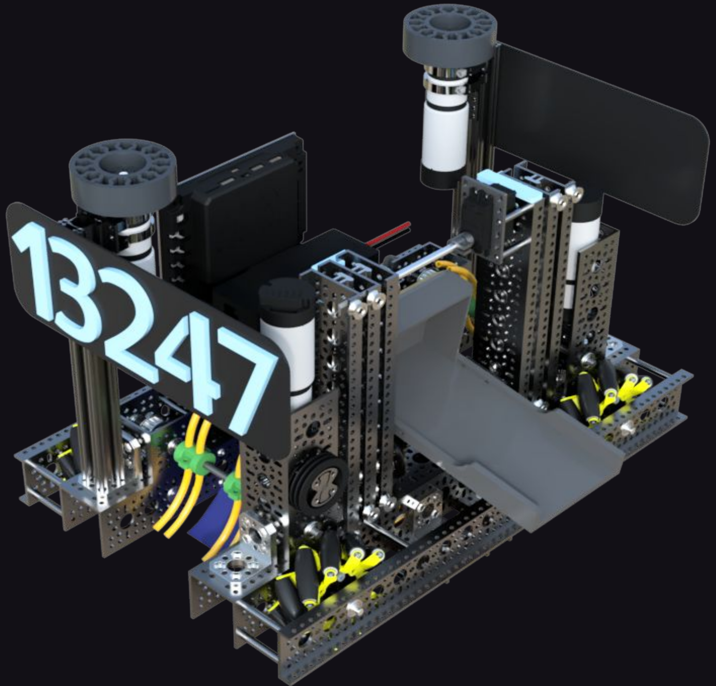
This season's challenge involved stacking cones on tall junction poles in different places through the field. The main arm for this design swung through the middle and allowed us to score cones from either side of our robot.

One challenge with this robot was scoring on both sides of the bot. Our solution was a 3D printed universal joint system embedded into our swinging arm to rotate the claw for scoring on both sides.

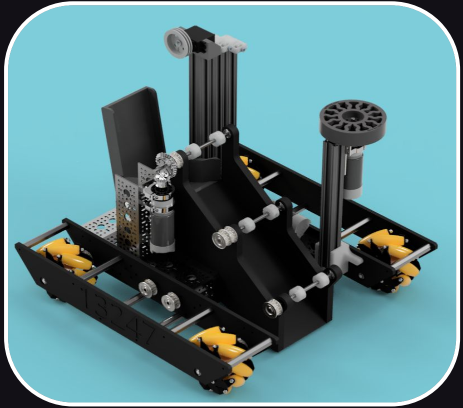
This claw used in this challenge used a 4 bar system to wrap around the game piece and secure it. The black TPU ends to the claw helped increase success in controlling game pieces.

2021-2022 Season Robot

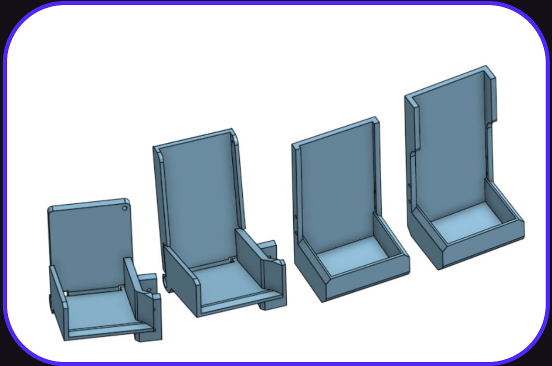
~Video~



The challenge for this season included picking up and placing blocks and balls at various heights. This design could intake game pieces from either side and outtake them through the middle.



In this challenge, we completely changed the design of our initial robot as seen here. The main concept was similar overall



This robot also was where our team involved our first ever 3D printed part to our bot. It was this bucket used to score game pieces.